

Intro to Sheaves

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- ▶ and if I restrict along the identity $U \subseteq U$, my group homomorphism should be the identity.

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- ▶ Slight lie – need to allow arbitrary unions $U = \bigcup V_\alpha$.
- ▶ Require also $F\emptyset = 0$ – some believe this follows from the above.

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- ▶ If X is a $\mathbb{C}$ -manifold, sheaf of *holomorphic* functions $\mathcal{O}_X$.
- ▶ Not all adjectives work. Nonexamples: bounded, compactly supported, etc.

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- ▶ Takeaway – continuous maps to a topological space Y form a sheaf!

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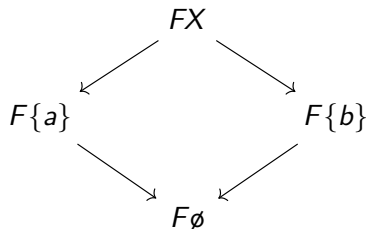
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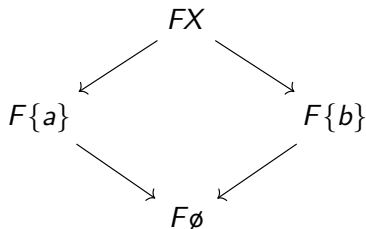
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- ▶ This square must commute – restricting from FX to $F\emptyset$ should be the same no matter what order you restrict.

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- ▶ Hence any pair in $Fa \times Fb$ gives rise to a unique $z \in FX$. These determine $z \in FX$.
- ▶ Conclusion: $FX = Fa \times Fb$.

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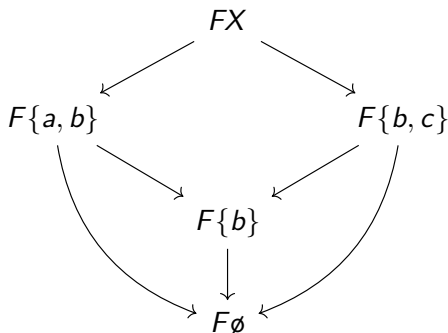
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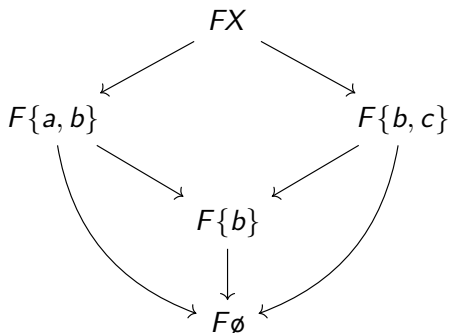
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- ▶ It doesn't matter what order I restrict

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- ▶ I.e., $FX = F\{a, b\} \times_{F\{b\}} F\{b, c\}$

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- ▶ The equivalence relation $\sim$ identifies $(s, U) \sim (t, V)$ if there's some $W \subseteq U \cap V$ such that $s|_W = t|_W$

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- ▶ E.g., sheaf of differentiable functions $\mathbb{R} \rightarrow \mathbb{R}$ – two functions induce the same stalk if they're the same in a neighborhood of p – they'll have the same limits and same derivatives at p .

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- ▶ For each p , we will get a map of stalks – $\varphi_p : F_p \rightarrow G_p$

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- ▶ Natural construction – think of least upper bound of a sequence, or the “ceiling” function that spits out the least integer greater than a fraction (“integerification”).

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$$f \circ g(x) = x$$

- ▶ Our sheafification will be a special subset of sections of the map $\text{Et } F \rightarrow X$.

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- ▶ “Evaluate” the section $s \in FU$ at the point p !

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- ▶ F^{sh} pieces together sections $s \in FU$ which agree locally!

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- ▶ Check : a section $t : U \rightarrow \text{Et } F$ is continuous iff t^{-1} of basis elements $\hat{s}(V)$ are open iff $t(p) = s_p$ on some open set V , for some $s \in FV$.

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- ▶ One can check whether a map $t : U \rightarrow Y$ is indeed a section locally – the $\Gamma(\cdot, f)$ is a *sheaf*!
- ▶ F^{sh} is $\Gamma(\cdot, \text{Et } F \rightarrow X)$, so F^{sh} is a sheaf!

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- ▶ A “local homeomorphism” is a map $f : Y \rightarrow X$ where there's an open cover $Y = \bigcup U_\alpha$ such that $f|_{U_\alpha} : U_\alpha \rightarrow f(U_\alpha)$ is a homeomorphism onto its image.